

Practical 4

AIM: To Implementation and Time analysis of factorial program using iterative and recursive method

1 ALGORITHM:

Algorithm 1:- factorial using iterative method

Input:- A non negative integer n.

Output:- The factorial of n, i.e. n!

Step 1:- start

Step 2:- Read the value of n

Step 3:- set fact = 1

Step 4:- for i = 1 to n, repeat:
fact = fact * i

Step 5:- display fact.

Step 6:- stop

Recursive method

Algorithm 2:- factorial using Recursive method

Input:- A non negative integer n.

Output:- The factorial of n, i.e. n!

Step 1:- start

Step 2:- Read value of n.

Step 3:- If n = 0 or n = 1, return:

Step 4:- Otherwise calculate:

$$\text{factorial}(n) = n \times \text{factorial}(n-1)$$

Step 5:- Return the calculated value.

Step 6:- display the factorial.

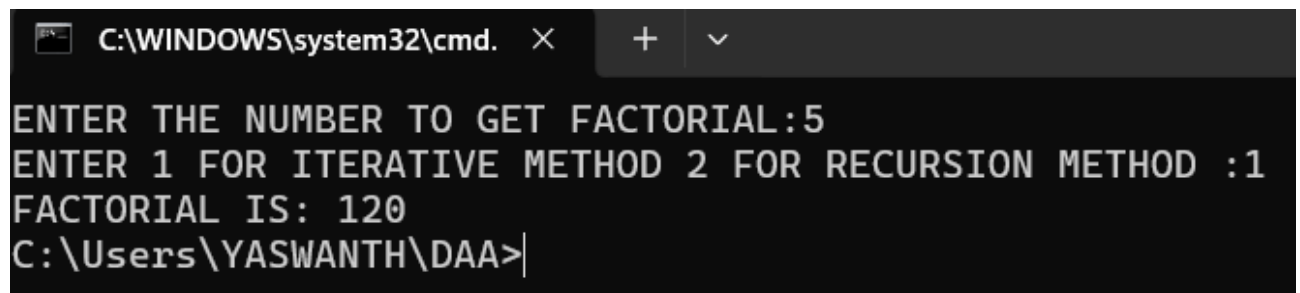
Step 7:- stop

2 PROGRAM:

```
#include<iostream>
using namespace std;
void it(int num){
    int fact=1;
    for(int i=1;i<=num;i++){
```

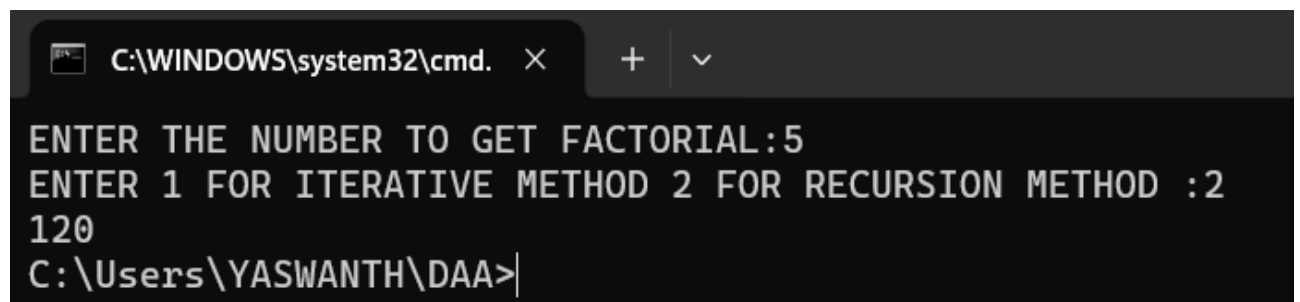
```
        fact*=i;
    }
    cout<<"FACTORIAL IS: "<<fact;
}
int re(int num){
    int fact=1;
    fact*=num;
    if(num==0){
        return 1;}
    return num*re(num-1);
}
int main(){
    int num;
    cout<<"ENTER THE NUMBER TO GET FACTORIAL:";
    cin>>num;
    int me;
    cout<<"ENTER 1 FOR ITERATIVE METHOD 2 FOR RECURSION METHOD :";
    cin>>me;
    if(me==1)
        it(num);
    else if(me==2)
        cout<<re(num);
    else{
        cout<<"ENTER THE CORRECT NUMBER";
    }
}
```

3 OUTPUT:



C:\WINDOWS\system32\cmd. X + v

ENTER THE NUMBER TO GET FACTORIAL:5
ENTER 1 FOR ITERATIVE METHOD 2 FOR RECURSION METHOD :1
FACTORIAL IS: 120
C:\Users\YASWANTH\DAA>



C:\WINDOWS\system32\cmd. X + v

ENTER THE NUMBER TO GET FACTORIAL:5
ENTER 1 FOR ITERATIVE METHOD 2 FOR RECURSION METHOD :2
120
C:\Users\YASWANTH\DAA>

4 TIME COMPLEXITY:

Time complexity :-

Iterative or functions :-

```
int fact = 1;
for (int i = 1; i <= n; i++) {
    fact = fact * i;
}
```

$$\Rightarrow T(n) = 1 + 1 + (n+1) + n + n$$

$$\text{Simplify} \Rightarrow T(n) = 3n + 3$$

$$\text{Therefore :- } (T(n) = 3n + 3)$$

ignoring constants & lower order terms :-

$$T(n) = O(n)$$

Time complexity :- Best $\geq$ Average $\geq$ worst $\geq O(n)$

For recursive method :-

$T(n)$ = time to compute factorial of n .

one multiplication $= O(1)$

one recursive call $= T(n-1)$

$$T(n) = T(n-1) + O(1)$$

$$\geq T(n-2) + 1 + 1$$

$$= T(n-3) + 1 + 1 + 1$$

$$= T(1) + (n-1)$$

$$T(n) = O(n)$$

5 Conclusion:

The factorial was successfully implemented using iterative and recursive methods. Both methods have $O(n)$ time complexity. The iterative method uses $O(1)$ space, while the recursive method uses $O(n)$ space.